



# Sex Classification from Human Scent Using Image Interpretation of 2D Gas Chromatography - Mass Spectrometry Data



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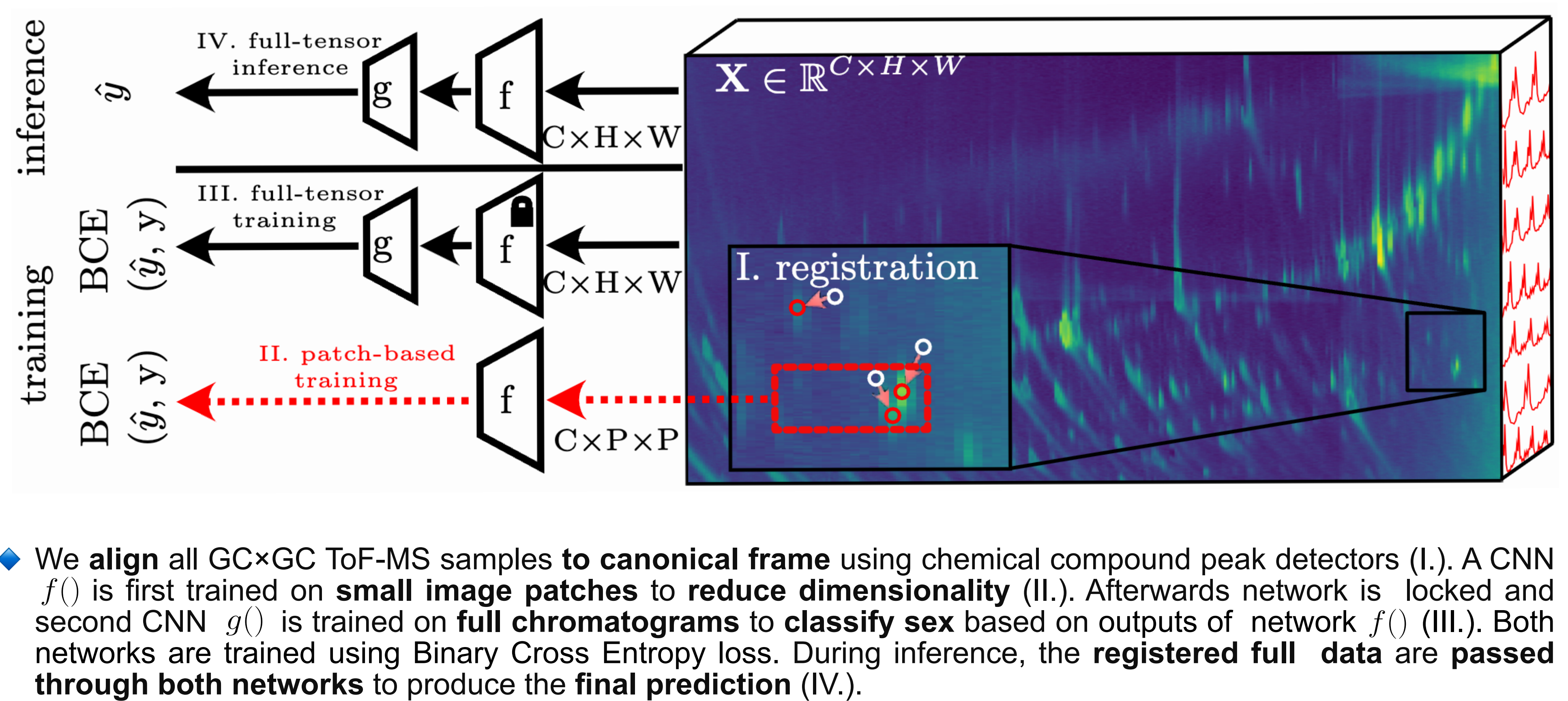


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## Introduction

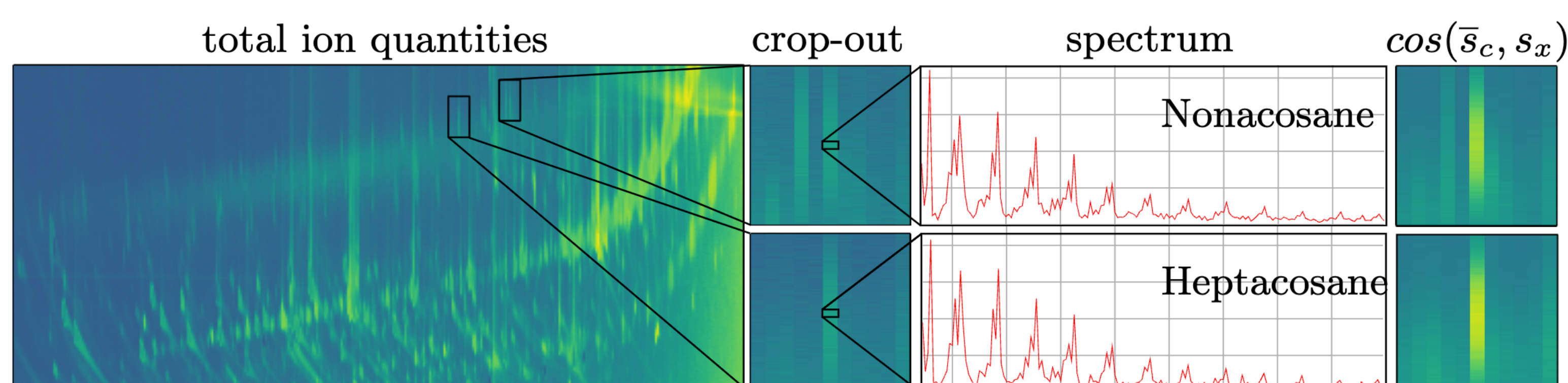
- ◆ **Two-dimensional Gas Chromatography with Time-of-Flight Mass Spectrometry** is an analytical technique that separates and identifies volatile compounds in complex mixtures, producing rich, high-dimensional chemical profiles. Human **scent** encodes subtle **biological** and identity-specific **information**, making it a promising source for sex classification.
- ◆ **Conventional chemical analysis** pipelines rely on **manual identification** of chemical compounds and their quantities. This is **labour-intensive** and subjective **process** which is **hardly reproducible**. To overcome these limitations, we treat raw **data as multi-channel images** and apply computer vision methods and machine learning for automated analysis.
- ◆ We **release a new dataset** specifically curated for this task. We also evaluate our method with **identity-aware cross-validation**. The proposed method achieves **~88% accuracy**, setting **new state-of-the-art** for sex classification from human scent samples.

## Method Overview



## I. Canonical Frame Registration

- ◆ Canonical frame registration is essential because both **compound retention times** and **chromatogram dimensions vary across measurements**, making direct comparison unreliable. To ensure that identical chemical compounds appear at consistent positions across samples, we align each measurement to canonical frame.



### 1. Peak Detection

- ◆ We estimate the highest-intensity pixel  $\hat{x}_c$  for each compound in  $|C|$  using compound peak detector.

### 2. Mesh Construction

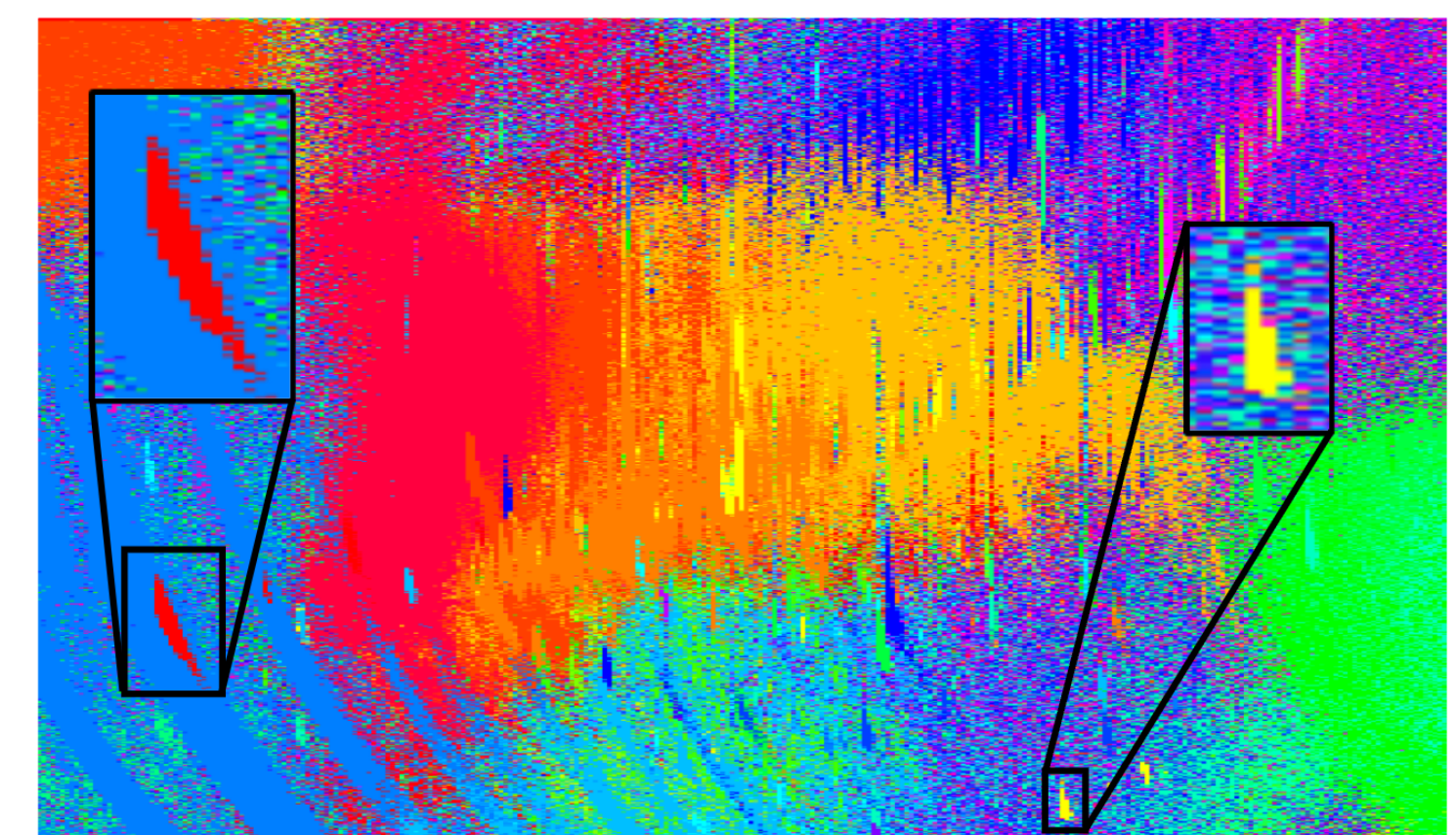
- ◆ We construct Delaunay triangulation from  $\{\hat{x}_c \mid c \in C\}$  to form a triangle mesh.

### 3. Coordinate Transformation

- ◆ We compute new coordinates within each triangle and align  $x_c$  with  $\hat{x}_c$  using piecewise linear interpolation.

### Compound Peak Detectors

- ◆ Cosine similarity based:
  - ◆  $\cos(\bar{s}_c, s)$  - cosine sim. between mass spectra
  - ◆  $c_{\cos}(\bar{x}_c, \bar{s}_c)$  - cosine sim. between mass spectra with additional penalty for long distance from expected spatial position
- ◆ Neural network based:
  - ◆ Uses mass spectra and spatial positions to classify each pixel into  $|C|$  classes.
  - ◆  $FCN(x, s)_{\text{pure}}$  - pixel with highest response for each compound  $c$
  - ◆  $FCN(x, s)_{\text{cos}}^+$  - combination of both highest response and cosine sim. for each chemical compound  $c$



## II. Sex Classification

- ◆ **Spatial coordinates are concatenated as channels** to preserve retention time information.
- ◆ We subsequently train **two neural networks**:
  - ◆  $f()$  - trained on patches of size  $18 \times 18$  to lower down the space
  - ◆  $g()$  - predicting the sex based on logits of  $f()$
- ◆ Patch-based learning acts as **both dimensionality reduction** and **data augmentation**.
- ◆ End-to-end training was infeasible due to data size and subtle spectral similarities.

## Data

- ◆ Each raw sample has shape  $\approx C \times 2000 \times 500$
- ◆ We evaluate on **two types of input data**:
  - ◆ **TIC** (Total Ion Current) - the sum over the  $C$  dimensions
  - ◆ **Full** - original size data

| Dataset | Num. Samples | Identities | Num. Samples per Identity |     |      |        |
|---------|--------------|------------|---------------------------|-----|------|--------|
|         |              |            | min                       | max | avg  | median |
| 4D-C    | 504          | 40         | 10                        | 37  | 12.6 | 10     |

## Experiments

### Compound Peak Detectors

- ◆ We evaluated **5 chemical compound detectors**.
- ◆ We used **5-fold cross-val.** with average error:

$$\frac{1}{k|C||M|} \sum_{i=1}^k \sum_{m \in M_i} \sum_{c \in C} |\hat{x}_c - x_c|$$

| Compound Classification Method   | AVG error $ \hat{x}_c - x_c $ (px) ↓ |                                  |
|----------------------------------|--------------------------------------|----------------------------------|
|                                  | BBox                                 | no BBox                          |
| $\cos(\bar{s}_c, s)$             | $21.4 \pm 4.1$                       | $63.5 \pm 5.1$                   |
| $c_{\cos}(\bar{x}_c, \bar{s}_c)$ | $20.1 \pm 3.8$                       | —                                |
| $FCN(x, s)_{\text{pure}}$        | $27.0 \pm 4.0$                       | $60.5 \pm 8.9$                   |
| $FCN(x, s)_{\text{cos}}$         | $20.7 \pm 4.2$                       | $29.1 \pm 2.1$                   |
| $FCN(x, s)_{\text{cos}}^+$       | $20.2 \pm 3.7$                       | <b><math>11.8 \pm 1.2</math></b> |

### Sex Classification

- ◆ We evaluated **TIC** and also **Full** data.
- ◆ We used **10-fold identity aware cross-val.** with average error:

$$\text{Err}(\mathcal{T}) = \frac{1}{k} \sum_{i=1}^k \frac{1}{|D_i|} \sum_{(x, y) \in D_i} \frac{1}{n_y} \mathbb{I}(g(f(x)) \neq y)$$

- ◆ We outperform SOTA by 14 percentage points

| Sex Classif. Model | peak detection                     |                            |  |
|--------------------|------------------------------------|----------------------------|--|
|                    | $c_{\cos}(\bar{x}_c, \bar{s}_c)$   | $FCN(x, s)_{\text{cos}}^+$ |  |
| TIC LinearCNN(X)   | $26.50 \pm 7.90$                   | $25.57 \pm 6.98$           |  |
| TIC CNN(X)         | $29.02 \pm 9.87$                   | $31.18 \pm 10.66$          |  |
| Full MaxPoolNN(X)  | $13.69 \pm 5.36$                   | $11.19 \pm 7.27$           |  |
| Full PatchNN(X)    | <b><math>11.60 \pm 4.72</math></b> | $11.75 \pm 5.76$           |  |